A Gram Decomposition for the Entropy Kernel in Liu's First Hypothesis

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Abstract

[HUMAN-AUDITED] We give a self-contained proof of the operational three-moment formulation of the positive-semidefiniteness hypothesis in Liu's conditionally-IID approach to the union-closed sets conjecture. For

$$K_{\mathrm{L}}(s,t) = h\big((1-s)(1-t) + s(1-s)t(1-t)\big), \qquad 0 \le s,t \le 1,$$

we prove that

$$\iint K_{\mathrm{L}}(s,t)\, d\mu(s)\, d\mu(t) \le 0$$

for every finite signed regular Borel measure μ annihilating 1, s, and $s(1-s)$. The reduction removes the projected rank-one terms and leaves a residual kernel R. An exact Taylor identity writes $-R$ as two positive-semidefinite power-series kernels plus an integral of reciprocal polynomial kernels. An explicit Lorentz-signature factorization makes every reciprocal kernel a convergent Gram series. [MACHINE-VERIFIED] The finite algebraic identities in this decomposition are also checked by exact symbolic assertions in the canonical repository script; those checks corroborate but do not replace the analytic proof. [OPEN] This manuscript is a candidate resolution of Liu's first hypothesis only. Liu's separate Hypothesis 2 and the stronger union-closed constant derived conditionally from both hypotheses remain open. A bounded theorem-level literature search found no earlier proof of this kernel statement; absence from that search is not proof of universal priority. No claim of peer review, independent acceptance, or end-to-end formal verification is made.

1 Introduction

[CITED-DEPENDENCY] A finite family of sets is *union-closed* if it is closed under pairwise unions. Frankl's union-closed sets conjecture asks whether every finite nontrivial union-closed family has an element belonging to at least half of its member sets; Bruhn and Schaudt survey the history and classical formulations [3].

[CITED-DEPENDENCY] Liu introduced a conditionally-IID entropy coupling and a finite-dimensional reduction for improving quantitative bounds toward that conjecture [6]. The explicit value reported in Liu's Theorem 13 is conditional on two distinct inputs: the positive-semidefiniteness condition in Section V-A and the asserted structure of a nine-parameter global minimizer in Section V-B. We call these inputs Hypothesis 1 and Hypothesis 2, respectively, following the repository audit terminology. Liu supported Section V-A with the 2,499-node

*Repository manuscript package: `math/LIU_H1/paper/`. Status labels distinguish human proof, machine-verified algebra, finite computation, and open scope.

grid code `frankl3.m` and finite coefficient code `frankl7.m`, and Section V-B with random-start optimization. Copies are archived under `../literature/code/`; all are numerical evidence rather than continuum/global proofs.

[HUMAN-AUDITED] The purpose of this paper is deliberately narrower than a new union-closed frequency theorem. We prove the operational formulation of Hypothesis 1, including all endpoints and all finite signed measures. The argument has three steps:

(i) the codimension-three projection is removed exactly;

(ii) the resulting residual kernel is written as a sum of three kernels;

(iii) the only non-obvious summand is made positive semidefinite by an explicit Lorentz-to-Gram conversion.

The residual statement is stronger than the projected assertion because it requires no moment constraints.

[HUMAN-AUDITED] Liu introduced the scale-one kernel and supported its conditional sign numerically. The closest rigorous predecessors are the unperturbed product-entropy kernel of Alweiss–Huang–Sellke [2, Lemma 5] and Liu's Lemma 11 for sufficiently small perturbation scale. Neither reaches $f(x) = x(1-x)$ at scale one. We are not aware of an earlier proof at that endpoint; the full multi-index search cutoff is 2026-08-25, an arXiv-feed refresh on 2026-08-26 found no later collision, and universal priority remains open.

The proof uses classical closure of PSD kernels under nonnegative sums, limits, and Schur products [4, Theorem 7.5.3]. Its reciprocal kernel is the standard Hilbert-ball kernel $(1-\langle u, v\rangle)^{-1}$ [1]. The problem-specific content is the exact entropy Taylor regrouping and the Lorentz factorization placing the normalized polynomial feature map inside the unit ball. Search protocol, sources, and limitations: `../LITERATURE_ORIGINALITY.md`.

Status vocabulary. Every load-bearing claim is assigned one of the following labels. [MACHINE-VERIFIED] means an exact symbolic or validated finite computation has a replayable code path. [HUMAN-AUDITED] means the continuum mathematical argument is written and checked as ordinary proof. [COMPUTATIONAL-EVIDENCE] means a finite-precision or finite-dimensional diagnostic only. [CITED-DEPENDENCY] identifies material imported from an external source. [OPEN] marks an unresolved question or incomplete audit, and [FAILED] marks a route for which an explicit obstruction is recorded. A machine-verified algebraic identity is not the same as an end-to-end formal proof in a proof assistant.

2 Spaces, kernels, and projectors

Let $X = [0,1]$ with its Borel σ-algebra. Write $\mathcal{M}(X)$ for the real vector space of finite signed regular Borel measures on X, equipped with total-variation norm, and write $L^2 = L^2(X, ds)$ for the real Hilbert space with Lebesgue measure. All unqualified integrals below are over $[0,1]$; double integrals are over $[0,1]^2$.

Define binary entropy in bits by

$$h(x) = -x\log_2 x - (1-x)\log_2(1-x), \qquad 0 < x < 1, \tag{1}$$

with the continuous endpoint values $h(0) = h(1) = 0$. Put

$$a(s) = s(1-s), \qquad z(s,t) = (1-s)(1-t) + a(s)a(t), \qquad K_{\mathrm{L}}(s,t) = h(z(s,t)). \tag{2}$$

The subscript L distinguishes Liu's entropy kernel from the auxiliary kernels introduced later. The functions z and K_{L} are symmetric and continuous on the closed square. Indeed, for fixed t, the function $(1-s)(1-t)(1+st)$ is nonincreasing in s, so $0 \le z \le 1$.

For a continuous symmetric kernel $F : X^2 \to \mathbb{R}$, define

$$\mathcal{Q}_F(\mu) = \iint F(s,t)\, d\mu(s)\, d\mu(t), \qquad \mu \in \mathcal{M}(X), \tag{3}$$

and define its bounded self-adjoint integral operator on L^2 by

$$(T_F g)(s) = \int F(s,t) g(t)\, dt. \tag{4}$$

We call F positive semidefinite (PSD) if every finite Gram matrix $[F(x_i,x_j)]_{i,j=1}^n$ is positive semidefinite; it is negative semidefinite (NSD) if $-F$ is PSD. Lemma 4.1 below connects this pointwise definition with (3).

Set

$$f_0(s) = 1, \qquad f_1(s) = s, \qquad f_2(s) = a(s), \qquad \mathcal{W} = \operatorname{span}\{f_0, f_1, f_2\} \subset L^2. \tag{5}$$

The algebraic identity $a(s) = s - s^2$, together with $s^2 = s - a(s)$, proves explicitly that

$$\mathcal{W} = \operatorname{span}\{1, s, s(1-s)\} = \operatorname{span}\{1, s, s^2\}. \tag{6}$$

Thus $\dim \mathcal{W} = 3$. The corresponding moment-annihilating spaces are

$$\mathcal{V}_{\mathcal{M}} = \left\{ \mu \in \mathcal{M}(X) : \int f_j\, d\mu = 0 \text{ for } j = 0,1,2 \right\}, \tag{7}$$

$$\mathcal{V}_2 = \mathcal{W}^{\perp} = \left\{ g \in L^2 : \int f_j(s) g(s)\, ds = 0 \text{ for } j = 0,1,2 \right\}. \tag{8}$$

For completeness, let

$$\boldsymbol{f}(s) = (f_0(s), f_1(s), f_2(s))^{\mathsf{T}}, \qquad \Gamma = \int \boldsymbol{f}(s)\boldsymbol{f}(s)^{\mathsf{T}}\, ds = \begin{pmatrix} 1 & 1/2 & 1/6 \\ 1/2 & 1/3 & 1/12 \\ 1/6 & 1/12 & 1/30 \end{pmatrix}.$$

Linear independence makes Γ invertible. The L^2-orthogonal projector onto $\mathcal{W}$ and the complementary projector onto $\mathcal{V}_2$ are

$$(\Pi_{\mathcal{W}} g)(s) = \boldsymbol{f}(s)^{\mathsf{T}} \Gamma^{-1} \int \boldsymbol{f}(t) g(t)\, dt, \qquad P_{\mathcal{V}} = I - \Pi_{\mathcal{W}}. \tag{9}$$

No other continuum projector is used in the proof.

The associated discrete projector can also be stated without ambiguity. For distinct nodes $s_1, \dots, s_n$ for which the $n \times 3$ design matrix C, $C_{ij} = f_{j-1}(s_i)$, has full column rank, put

$$P_n = I_n - C(C^{\mathsf{T}} C)^{-1} C^{\mathsf{T}}. \tag{10}$$

Then P_n is the Euclidean orthogonal projector onto $\ker C^{\mathsf{T}}$, a subspace of dimension $n-3$.

Hypothesis 2.1 (Operational form of Liu's Hypothesis 1; [CITED-DEPENDENCY])**.** For every $\mu \in \mathcal{V}_{\mathcal{M}}$,

$$\mathcal{Q}_{K_{\mathrm{L}}}(\mu) = \iint K_{\mathrm{L}}(s,t)\, d\mu(s)\, d\mu(t) \le 0. \tag{11}$$

Equivalently at the level of the named constraints, μ annihilates $1, s, s(1-s)$, or, by (6), $1, s, s^2$.

Remark 2.2 (The source's codimension count). [CITED-DEPENDENCY] Liu's Section V-A calls the affine probability-measure slice "codimension 2" [6, Section V-A]: normalization already fixes mass, and equations (48)–(49) add two moments. Homogeneous signed perturbations must also annihilate mass, so the accompanying numerical projector has the three columns $1, s, s(1-s)$. By (6) these are the independent moments $1, s, s^2$. Thus "codimension two in the probability simplex" and "three homogeneous constraints on signed measures" describe the same operational tangent space.

Theorem 2.3 (Unprojected residual theorem; [HUMAN-AUDITED])**.** *Define the continuous symmetric kernel R by*

$$R(s,t) = A\big[B - (1+B)\ln(1+B)\big] - \big[z + (1-z)\ln(1-z)\big], \tag{12}$$

where

$$u = 1-s, \qquad v = 1-t, \qquad A = uv, \qquad B = st, \qquad z = A(1+B). \tag{13}$$

The convention at $z = 1$ is the continuous one. Then R is NSD on the whole closed interval: for every $\nu \in \mathcal{M}(X)$,

$$\mathcal{Q}_R(\nu) = \iint R(s,t)\, d\nu(s)\, d\nu(t) \le 0. \tag{14}$$

No moment, mass, sign, absolute-continuity, or endpoint-exclusion assumption is placed on ν.

Corollary 2.4 (Candidate resolution of Liu's Hypothesis 1; [HUMAN-AUDITED])**.** *Hypothesis 2.1 holds. In particular,*

$$\langle g, T_{K_{\mathrm{L}}} g\rangle_{L^2} \le 0 \quad (g \in \mathcal{V}_2), \qquad -P_{\mathcal{V}} T_{K_{\mathrm{L}}} P_{\mathcal{V}} \succeq 0 \quad \text{on } L^2. \tag{15}$$

For every discrete node set covered by (10), $-P_n[K_{\mathrm{L}}(s_i,s_j)]_{i,j}P_n$ is positive semidefinite.

Theorem 2.3 and Corollary 2.4 are proved in Sections 3–6.

3 Exact removal of the projector

Use natural logarithms in this section and define

$$G(x) = x + (1-x)\ln(1-x), \qquad -1 \le x < 1, \quad G(1) = 1. \tag{16}$$

On $[0,1]$,

$$G(x) = \sum_{k=2}^{\infty} \frac{x^k}{k(k-1)}. \tag{17}$$

Lemma 3.1 (Projected entropy equals the residual form; [HUMAN-AUDITED])**.** *For every $\mu \in \mathcal{V}_{\mathcal{M}}$,*

$$(\ln 2)\, \mathcal{Q}_{K_{\mathrm{L}}}(\mu) = \mathcal{Q}_R(\mu). \tag{18}$$

All terms are interpreted by continuous extension at $s = 0, 1$ and $t = 0, 1$.

Proof. From $z = uv(1+st)$ and (1),

$$(\ln 2) h(z) = -z\ln u - z\ln v - z\ln(1+B) + z - G(z). \tag{19}$$

Here, for example, $u \ln u$ is set to zero at $u = 0$; every apparently singular product in (19) then has a continuous value. Since $z = A + AB$, move the rank-one term $AB = a(s)a(t)$ into the remainder. The result is the pointwise identity

$$(\ln 2) h(z) = -z\ln u - z\ln v + A + R(s,t). \tag{20}$$

Indeed, the remainder on the right is exactly (12).

It remains to show that the first three kernels on the right of (20) vanish in the quadratic form of μ. Their rank-one decompositions are

$$\begin{aligned} -z\ln u &= -(u\ln u)v - (us\ln u)(vt), \\ -z\ln v &= -u(v\ln v) - (us)(vt\ln v), \\ A &= uv. \end{aligned}$$

The functions $u = 1 - s$ and $us = s(1-s)$ belong to $\mathcal{W}$; the analogous statements hold in the t-variable. Each rank-one product therefore has at least one factor annihilated by μ. Fubini's theorem [8, Chapter 8] applies because all extended factors are continuous and μ has finite total variation. Integrating (20) proves (18). □

4 Positive-semidefinite closure and the Taylor identity

Lemma 4.1 (Closure rules for continuous PSD kernels; [HUMAN-AUDITED]). *Let Y be a compact metric space.*

(a) A rank-one kernel $q(x)q(y)$ is PSD.

(b) A nonnegative linear combination, a pointwise limit, or a pointwise product of finite-valued PSD kernels is PSD whenever the resulting kernel is finite. A finite nonnegative integral of PSD kernels is PSD whenever its entries are measurable and integrable.

(c) If F is continuous and PSD on Y^2, then $\iint F\,d\nu\,d\nu \geq 0$ for every $\nu \in \mathcal{M}(Y)$.

Proof. Part (a) follows from $\sum_{i,j} c_i c_j q(x_i)q(x_j) = (\sum_i c_i q(x_i))^2$. For part (b), nonnegative sums and integrals preserve each finite Gram quadratic form. Pointwise limits preserve positive semidefiniteness because the cone of positive semidefinite finite matrices is closed. Products are covered by the Schur product theorem [4, Theorem 7.5.3]; equivalently, Gram vectors may be tensor-multiplied.

For part (c), partition Y into finitely many Borel sets of mesh tending to zero and choose one point from each nonempty cell. Replacing ν by the atomic measure that assigns each chosen point the signed mass of its cell produces nonnegative quadratic forms. Uniform continuity of F bounds the difference from the original double integral by the oscillation of F on the product cells times $|\nu|(Y)^2$, which tends to zero. The limit is therefore nonnegative. □

For $0 \leq r \leq 1$, define the polynomial kernel

$$D_r(s,t) = (1 + rB)\,[1 - A(1 + rB)]. \tag{21}$$

It is positive away from $(s,t) = (0,0)$: the first factor is positive, while $A(1 + rB) \leq A(1 + B) = z \leq 1$, with equality only at the corner.

Lemma 4.2 (Exact Taylor decomposition; [HUMAN-AUDITED] and [MACHINE-VERIFIED]). *For every $(s,t) \in [0,1]^2 \setminus \{(0,0)\}$,*

$$\begin{aligned} -R(s,t) = G(A) &- AB\ln(1-A) \\ &+ B^2 \int_0^1 (1-r)\frac{A}{D_r(s,t)}\,dr. \end{aligned} \tag{22}$$

At $(s,t) = (0,0)$, assigning the three terms on the right the values $1, 0, 0$, respectively, gives the same identity. These assignments are proved continuous in (27) and Lemma 5.2.

Proof. Because $G(-B) = -B + (1+B)\ln(1+B)$, equation (12) gives

$$-R = G(A(1+B)) + AG(-B). \tag{23}$$

For fixed $0 \le A < 1$, let the right-hand side be $\phi_A(B)$. Direct differentiation yields, for all B in the relevant interval,

$$\phi_A(0) = G(A), \qquad \phi_A'(0) = -A\ln(1-A), \qquad \phi_A''(B) = \frac{A}{(1+B)[1-A(1+B)]}. \tag{24}$$

Twice applying the fundamental theorem of calculus gives

$$\phi_A(B) - \phi_A(0) - B\phi_A'(0) = \int_0^B (B-x)\phi_A''(x)\,dx = B^2\int_0^1 (1-r)\phi_A''(rB)\,dr.$$

This is (22). The excluded value $A = 1$ occurs only at $(s,t) = (0,0)$, where $B = 0$. Directly from (12), $R(0,0) = -1$, so the assigned endpoint values also give the identity there. □

The first two kernels in (22) are already PSD. Indeed, by (17),

$$G(A) = \sum_{k=2}^{\infty} \frac{u^k v^k}{k(k-1)}, \tag{25}$$

$$-AB\ln(1-A) = \sum_{k=1}^{\infty} \frac{A^{k+1}B}{k} = \sum_{k=1}^{\infty} \frac{(u^{k+1}s)(v^{k+1}t)}{k}. \tag{26}$$

Every summand is rank one with nonnegative coefficient. At the sole corner where the logarithmic expression is indeterminate, if $m = \max\{s,t\}$, then

$$0 \le AB[-\ln(1-A)] \le m^2[-\ln m] \longrightarrow 0, \tag{27}$$

because $1 - A = s + t - st \ge m$ and $B = st \le m^2$. Hence the pointwise-limit rule in Lemma 4.1 applies on the closed square.

5 Lorentz factorization and reciprocal Gram kernels

The remaining integrand in (22) contains the kernel AB^2/D_r. The numerator is rank one:

$$AB^2 = (1-s)s^2(1-t)t^2 = q(s)q(t), \qquad q(s) = (1-s)s^2. \tag{28}$$

The reciprocal denominator is handled next.

Lemma 5.1 (Exact Lorentz factorization; [HUMAN-AUDITED] and [MACHINE-VERIFIED])**.** *For* $0 \le r \le 1$*, define*

$$\zeta_0(s) = -\frac{2rs^2 - (r+1)s + 1}{r+1}, \qquad \zeta_1(s) = 1 + 2rs^2,$$

$$\zeta_2(s) = \frac{s^2(r+2-rs)}{r+2}, \qquad \zeta_3(s) = s^3,$$

and

$$\alpha_r(s) = \frac{\zeta_1(s)}{\sqrt{r+1}}, \qquad \boldsymbol{b}_r(s) = \left(\sqrt{r+1}\,\zeta_0(s), \sqrt{r(r+2)}\,\zeta_2(s), r\sqrt{\frac{2}{r+2}}\,\zeta_3(s)\right). \tag{29}$$

Then, for all $s, t \in [0,1]$,

$$D_r(s,t) = \alpha_r(s)\alpha_r(t) - \langle \boldsymbol{b}_r(s), \boldsymbol{b}_r(t) \rangle_{\mathbb{R}^3}. \tag{30}$$

Moreover,

$$D_r(s,s) \geq s(2 - 2s + 2s^2 - s^3) > 0 \qquad (0 < s \leq 1). \tag{31}$$

Proof. With $\boldsymbol{m}(s) = (1, s, s^2, s^3)^\mathsf{T}$, direct coefficient collection in (21) gives

$$D_r(s,t) = \boldsymbol{m}(s)^\mathsf{T} M(r) \boldsymbol{m}(t), \quad M(r) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & -r-1 & 2r & 0 \\ 0 & 2r & -r(r+2) & r^2 \\ 0 & 0 & r^2 & -r^2 \end{pmatrix}. \tag{32}$$

Let S be the permutation matrix that swaps the first two coordinates and put $\widetilde{\boldsymbol{m}}(s) = (s, 1, s^2, s^3)^\mathsf{T}$. Exact multiplication gives

$$S^\mathsf{T} M(r) S = L(r)\Delta(r)L(r)^\mathsf{T}, \tag{33}$$

where

$$L(r) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -\frac{1}{r+1} & 1 & 0 & 0 \\ -\frac{2r}{r+1} & 2r & 1 & 0 \\ 0 & 0 & -\frac{r}{r+2} & 1 \end{pmatrix}, \qquad \Delta(r) = \operatorname{diag}\left(-r-1, \frac{1}{r+1}, -r(r+2), -\frac{2r^2}{r+2}\right).$$

A second exact multiplication gives

$$L(r)^\mathsf{T} \widetilde{\boldsymbol{m}}(s) = (\zeta_0(s), \zeta_1(s), \zeta_2(s), \zeta_3(s))^\mathsf{T}.$$

Substitution into (33) is precisely (30), including $r = 0$, where two negative coordinates vanish.

For the diagonal bound, set $q_r = 1 + rs^2$. Then

$$D_r(s,s) = q_r[1 - (1-s)^2 q_r], \qquad 1 \leq q_r \leq 1 + s^2.$$

The displayed quadratic is concave as a function of q_r, so its minimum on this interval occurs at an endpoint. At the endpoints,

$$1 - (1-s)^2 = s(2-s),$$
$$1 - (1-s)^2(1+s^2) = s(2 - 2s + 2s^2 - s^3).$$

The first endpoint value is at least the right side of (31), because $(2-s) - (2 - 2s + 2s^2 - s^3) = s(1-s)^2 \geq 0$; the second endpoint has the additional factor $1 + s^2 \geq 1$. Finally, the cubic $p(s) = 2 - 2s + 2s^2 - s^3$ decreases from 2 to 1 on $[0,1]$: its derivative has discriminant -8 and is negative at $s = 0$. This proves (31). ☐

Lemma 5.2 (Reciprocal Gram series, including the boundary; [HUMAN-AUDITED])**.** *For each* $r \in [0,1]$*, define*

$$H_r(s,t) = \begin{cases} \dfrac{AB^2}{D_r(s,t)}, & (s,t) \neq (0,0), \\ 0, & (s,t) = (0,0). \end{cases} \tag{34}$$

Then H_r *is a continuous PSD kernel on* $[0,1]^2$*. The continuity at the corner is uniform in* $r \in [0,1]$*.*

Proof. The function $\alpha_r(s) > 0$ on $[0,1]$. For $s > 0$, put $\boldsymbol{c}_r(s) = \boldsymbol{b}_r(s)/\alpha_r(s)$. From Lemma 5.1,

$$1 - \|\boldsymbol{c}_r(s)\|^2 = \frac{D_r(s,s)}{\alpha_r(s)^2} > 0.$$

Consequently, for $s, t > 0$, Cauchy–Schwarz gives $|\langle \boldsymbol{c}_r(s), \boldsymbol{c}_r(t)\rangle| < 1$, and the geometric series yields the absolutely convergent identity

$$\frac{1}{D_r(s,t)} = \frac{1}{\alpha_r(s)\alpha_r(t)} \sum_{k=0}^{\infty} \left\langle \boldsymbol{c}_r(s)^{\otimes k}, \boldsymbol{c}_r(t)^{\otimes k} \right\rangle, \tag{35}$$

where the $k = 0$ inner product is 1. Each summand is a Gram kernel. Thus $1/D_r$ is PSD on $(0,1]^2$, and multiplying its feature vectors by $q(s) = (1-s)s^2$ shows that H_r is PSD there.

If exactly one of s, t is zero, the denominator is positive and the numerator is zero, so (34) is continuous with value zero. At the remaining corner, set $E_r = 1 - A(1 + rB)$. Exact factorization gives

$$\begin{aligned} E_r - s &= (1-s)t\,[1 - rs(1-t)] \geq 0, \\ E_r - t &= (1-t)s\,[1 - rt(1-s)] \geq 0. \end{aligned} \tag{36}$$

Thus $D_r = (1 + rB)E_r \geq \max\{s,t\} =: m$, uniformly for $r \in [0,1]$. Since $0 \leq AB^2 \leq s^2t^2 \leq m^4$,

$$0 \leq H_r(s,t) \leq m^3 \longrightarrow 0 \qquad \text{as } (s,t) \to (0,0),$$

again uniformly in r. For $\varepsilon \in (0,1)$, define

$$H_{r,\varepsilon}(s,t) = H_r\big(\max\{s,\varepsilon\}, \max\{t,\varepsilon\}\big).$$

This is the pullback of the PSD kernel H_r on $[\varepsilon,1]^2$, hence is PSD on the whole closed square. The continuity just proved gives $H_{r,\varepsilon} \to H_r$ pointwise as $\varepsilon \downarrow 0$, so Lemma 4.1 proves that the extended H_r is PSD. □

6 Assembly of the proof

Proof of Theorem 2.3. By Lemma 4.2,

$$-R = G(A) - AB\ln(1-A) + \int_0^1 (1-r)H_r\,dr. \tag{37}$$

The first kernel is PSD by (25) and the second by (26), with its closed-square extension justified by (27). Lemma 5.2 proves that every H_r is continuous and PSD. Away from $(s,t) = (0,0)$, the map $(r,s,t) \mapsto H_r(s,t)$ is rational with positive denominator; the uniform bound $0 \leq H_r(s,t) \leq \max\{s,t\}^3$ extends it continuously at the corner. It is therefore bounded and jointly continuous on $[0,1]^3$. In particular, every entry $r \mapsto H_r(x_i,x_j)$ is measurable and integrable, and integration in r preserves continuity in (s,t). Since $1 - r \geq 0$, the integral in (37) is PSD by Lemma 4.1. Hence $-R$ is a continuous PSD kernel on $[0,1]^2$. Applying part (c) of that lemma to an arbitrary $\nu \in \mathcal{M}(X)$ gives $-\mathcal{Q}_R(\nu) \geq 0$, which is (14). □

Proof of Corollary 2.4. For $\mu \in \mathcal{V}_{\mathcal{M}}$, Lemma 3.1 and Theorem 2.3 give

$$(\ln 2)\mathcal{Q}_{K_{\mathrm{L}}}(\mu) = \mathcal{Q}_R(\mu) \leq 0.$$

Since $\ln 2 > 0$, this is Hypothesis 2.1.

If $g \in \mathcal{V}_2$, the signed measure $d\mu(s) = g(s)\,ds$ is finite because $[0,1]$ has finite measure and $L^2 \subset L^1$. It belongs to $\mathcal{V}_{\mathcal{M}}$, so $\langle g, T_{K_{\mathrm{L}}} g\rangle = \mathcal{Q}_{K_{\mathrm{L}}}(\mu) \le 0$. For arbitrary $g \in L^2$, apply this inequality to $P_{\mathcal{V}} g$ to obtain the operator statement in (15).

Finally, for nodes s_i and a vector $c \in \ker C^{\mathsf{T}}$, the atomic measure $\sum_i c_i \delta_{s_i}$ lies in $\mathcal{V}_{\mathcal{M}}$. Thus $c^{\mathsf{T}}[K_{\mathrm{L}}(s_i, s_j)]c \le 0$. Since P_n maps onto $\ker C^{\mathsf{T}}$, this is exactly the asserted positive semidefiniteness of the negative projected matrix. □

7 Discussion and limits of the result

[HUMAN-AUDITED] The proof establishes a stronger unprojected statement for R, then transfers it to the projected entropy kernel by an exact identity. No grid limit, spectral truncation, unproved exchange of infinite sums, or assumption about the support of the signed measure is used. Three infinite Gram series appear: (25), (26), and (35). The first converges uniformly because its nonnegative coefficients sum to one. The second and third converge absolutely away from their singular corner and extend to the closed square by (27) and the uniform $H_r \le \max\{s,t\}^3$ estimate, respectively.

[OPEN] Liu's Hypothesis 2 is a separate global optimization assertion for a nine-parameter objective. Nothing in Theorem 2.3 proves that assertion, identifies its global minimizers, or validates the stronger union-closed constant that Liu derives when both hypotheses are assumed. Accordingly, both Liu's printed decimal 0.382709087918741 and the nearby equation-defined value discussed in the canonical audit remain conditional as union-closed frequency claims. This paper does not claim an unconditional result at either value, nor does it claim Frankl's conjectured constant 1/2.

[HUMAN-AUDITED] The completed bounded literature audit found this proof apparently new through its 2026-08-25 cutoff, with no citation claiming or proving Liu's Hypothesis 1. That is a search result, not a universal priority theorem. The result remains a candidate until ordinary external mathematical review; no statement here implies peer review or journal acceptance.

[HUMAN-AUDITED] The continuum proof is ordinary mathematics, not a formal proof-assistant development. [MACHINE-VERIFIED] labels below refer only to specified exact algebraic assertions or validated finite computations. Therefore the package does not claim end-to-end formal verification.

A Computational verification and separation of roles

The analytic proof in Sections 3–6 is load-bearing. The computations in this appendix have narrower roles. Paths are relative to this manuscript directory unless explicitly stated otherwise.

Artifact: `../verification/independent_exact_checker.py`. [MACHINE-VERIFIED] CAS-free coefficient checks over exact rational arithmetic cover the reduction, projector span, Lorentz factorization, diagonal bound, and corner factors.

Artifact: `../../uc/liu_R_nsd.py`. [MACHINE-VERIFIED] A clean isolated run completed successfully and checked the canonical SymPy identities, finite Arb compression bounds, and floating diagnostics. Its exact stdout is `../verification/logs/liu_R_nsd.clean.log`; the analytic continuum conclusion remains [HUMAN-AUDITED].

Artifact: `../../uc/liu_kernel.py`. [COMPUTATIONAL-EVIDENCE] This reproduces the projected grid calculation and the operational three-column projector; it is not a continuum proof.

Artifact: `../../uc/liu_moment_cert.py`. [MACHINE-VERIFIED] Its exact finite-moment, finite-node Arb, and interval Cholesky claims replayed successfully; the log is `../verification/lo` gs/liu_moment_cert.clean.log. [COMPUTATIONAL-EVIDENCE] Its floating spectra remain finite-dimensional corroboration only.

Artifact: ../../uc/liu_tail.py. [FAILED] This records obstructions to earlier tail-domination routes. [COMPUTATIONAL-EVIDENCE] Its sampled diagnostics are not imported into the proof above.

A.1 Exact symbolic checks

[MACHINE-VERIFIED] Two separate exact code paths audit the finite algebraic core. The function `exact_factorisation_checks()` in `../../uc/liu_R_nsd.py` uses SymPy exact expressions [7]. The package-local `../verification/independent_exact_checker.py` uses only Python integers and `fractions.Fraction`; it does not import the canonical module or any computer-algebra, numerical, or interval package. Between them, the exact assertions cover:

(1) the polynomial reduction $z = A(1 + B)$, the removal $z - AB = A$, and the full rank of $\{1, s, s(1 - s)\}$;

(2) the three values in (24);

(3) coefficient extraction of D_r into $M(r)$, its determinant, and its exact characteristic polynomial;

(4) the LDL^{T} and cleared-denominator versions of the Lorentz factorization, including $r = 0$;

(5) the diagonal polynomial and nonnegative-factor certificates used in (31);

(6) both exact corner identities in (36).

Successful replay checks transcription and algebra. Neither code path verifies Taylor's theorem, logarithm differentiation, kernel closure, convergence, Fubini's theorem, signed-measure approximation, or the logical assembly of the result; those are the [HUMAN-AUDITED] arguments in the body.

A.2 Finite validated and numerical corroboration

[MACHINE-VERIFIED] The clean isolated canonical run completed successfully at the pinned environment in `../REPRODUCIBILITY.md`. The degree $8, 12, 16$ computations give certified negative one-sided upper bounds for compressed R eigenvalues through a finite Loewner majorant. They do not prove the continuum theorem. [COMPUTATIONAL-EVIDENCE] The same run reconstructs (22) at quadrature nodes and inspects sampled eigenvalues; these are regression tests only.

From the `math/` directory, run the independent exact checker with

```
python3-I-BLIU_H1/verification/independent_exact_checker.py.
```

The standard-library-only checker and every canonical clean command, exact package version, source/log hash, observed exit status, and runtime are recorded in `../REPRODUCIBILITY.md`. The canonical entry point is executed in an isolated `uv` environment rather than trusting the repository virtual environment. Exit status zero and the complete labeled output are required.

A.3 Trust boundary

The computational trust boundary includes the invoked Python interpreter, SymPy for exact expression manipulation, and, for interval subroutines, the Python-FLINT/Arb implementation [5]. Floating diagnostics additionally depend on NumPy and SciPy. No numerical result is required for the continuum proof. The manuscript does not assert that these tools constitute a proof assistant or an independently certified implementation.

B Reproducibility checklist

A reproducibility review should distinguish the following checks.

R1. [HUMAN-AUDITED] Verify the definitions (1)–(13), including the base-two versus natural-log factor $\ln 2$.

R2. [HUMAN-AUDITED] Verify the span identity (6) before comparing the source's codimension terminology with its operational projector.

R3. [HUMAN-AUDITED] Check every rank-one factor in (20); this is the entire use of the three moment constraints.

R4. [MACHINE-VERIFIED] Replay the exact symbolic assertions and compare them with (24), (32), and (33).

R5. [HUMAN-AUDITED] Check positivity of $D_r(s, s)$, absolute convergence in (35), and the uniform corner estimate in Lemma 5.2.

R6. [HUMAN-AUDITED] Check the atomic approximation in Lemma 4.1; this is what extends finite Gram positivity to every finite signed regular Borel measure.

R7. [HUMAN-AUDITED] Read the bounded literature-search result separately from mathematical correctness. Its 2026-08-25 cutoff and inaccessible sources prohibit an unqualified novelty or priority claim.

The LaTeX manuscript itself can be built from this directory with the standard sequence

`pdflatexmain.tex  bibtexmain  pdflatexmain.tex  pdflatexmain.tex`.

A successful typesetting build checks cross-reference and bibliography syntax; it does not verify the theorem.

C Data and code availability statement

No empirical dataset is used in the theorem. The source of truth for the proof computations remains the canonical code under `../../uc/`; this paper does not duplicate or fork those scripts. Package-specific replay records belong under `../verification/`. The manuscript sources are `main.tex` and `refs.bib`. Availability and reuse are governed by the repository's existing terms; this paper does not assert an additional software or data license.

[HUMAN-AUDITED] The literature and originality audit at `../LITERATURE_ORIGINALITY.md` is complete through its 2026-08-25 cutoff. It records three indexed sources as inaccessible and excludes private or unindexed work; accordingly this paper makes no unqualified priority claim. Hypothesis 2 code and experiments elsewhere in the repository are outside this paper's theorem and data claim.

Acknowledgment of scope

This manuscript isolates one kernel theorem from a larger union-closed audit. The status labels are intended to keep analytic proof, symbolic checking, numerical corroboration, cited context, failed routes, and open questions logically separate.

References

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